Elective-I

UEI612: ROBOTICS

L	T	P	Cr
2	0	2	3.0

Course Objectives: To introduce the concepts of Robotic system, its components and instrumentation and control related to robotics.

Basic Concepts in Robotics: Automation and robotics, Robot anatomy, Basic structure of robots, Resolution, Accuracy and repeatability.

Classification and Structure of Robotic System: Point to point and continuous path systems. Control loops of robotic systems, The manipulators, The wrist motion and grippers.

Drives and Control Systems: Hydraulic systems, Dc servo motors, Basic control systems concepts and models, Control system analysis, Robot activation and feedback components. Positional and velocity sensors, actuators. Power transmission systems, Robot joint control design.

Robot arm Kinematics and Dynamics: The direct kinematics problem, The inverse kinematics solution, Lagrange-Euler formation, Generalized D'Alembert equations of motion, Denavit Hartenberg convention and its applications.

Motion planning and control Joint and Cartesian space trajectory planning and generation, Classical control concepts using the example of control of a single link, Independent joint PID control, Control of a multi-link manipulator,

Robotic Sensors: Sensors and Instrumentation in robotics: Tactile sensors, proximity and range sensors, Force and torque sensors, Uses of sensors in robotics. Vision Systems: Vision equipment, Image processing, Concept of low level and high level vision.

Course Learning Outcomes (CLO):

Student will be able to:

1. Demonstrate the basic concepts of robotics, their classification and structure
2. Exhibit the knowledge of type of the drive and control systems used in robotics
3. Design the kinematics and dynamics of robotic arm for motion control
4. Describe the type of sensors and other instruments used in robotics

Text Books:

1. *Nikku, S.B., Introduction to Robotics, Prentice Hall of India Private-2002*
2. *Schilling. R. J., Fundamentals of Robotics: Analysis and Control, Prentice Limited (2006)*

Reference Books:

1. *Criag, J., Fundamentals of Robotics: Analysis and Control, Prentice Limited (2006).*

Evaluation Scheme:

Sr. No.	**Evaluation Elements**	**Weightage (%)**
1	MST+EST	70
2	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30